

Drones: 2 | Iterations: 100

Connection using udp:// is deprecated, please use udpin:// or udpout:// (cli_arg.cpp:28)

[Port 14540] Waiting for connection...

[Port 14540] Connected!

Connection using udp:// is deprecated, please use udpin:// or udpout:// (cli_arg.cpp:28)

[Port 14541] Waiting for connection...

[Port 14541] Connected!

[Port 14540] Waiting for GPS/sensor readiness...

[Port 14540] GPS ready.

[Port 14540] Waiting for PX4 pre-arm checks to settle...

[Port 14540] Arming...

[Port 14540] Armed successfully.

[Port 14540] Offboard mode started.

[Port 14540] Taking off...

[Port 14540] At altitude. Ready for algorithm control.

[Port 14541] Waiting for GPS/sensor readiness...

[Port 14541] GPS ready.

[Port 14541] Waiting for PX4 pre-arm checks to settle...

[Port 14541] Arming...

[Port 14541] Armed successfully.

[Port 14541] Offboard mode started.

[Port 14541] Taking off...

[Port 14541] At altitude. Ready for algorithm control.

[LiDAR 0] Subscribed to /world/windy/model/x500_0/link/lidar_sensor_link/sensor/lidar/scan

[LiDAR 1] Subscribed to /world/windy/model/x500_1/link/lidar_sensor_link/sensor/lidar/scan

Sensor subscribers started. Waiting 2s to connect...

LiDAR[0] min_range=infm clear

LiDAR[1] min_range=infm clear

Drone 0 raw position: N=-0.005 E=-0.004 Alt=15.010

Drone 1 raw position: N=0.005 E=0.024 Alt=14.956

LiDAR[0] min_range=infm clear

LiDAR[1] min_range=infm clear

=== Starting algorithm loop ===

--- Iteration 1/100 ---

Best fitness so far: 56.548

x_best: [5.26071200e-03 2.44975947e-02 1.49563065e+01]

New x_best found: [29.42812819 10.89410412 18.22962775]

--- Iteration 2/100 ---

Best fitness so far: 31.134

x_best: [29.42812819 10.89410412 18.22962775]

--- Iteration 3/100 ---

Best fitness so far: 31.134

x_best: [29.42812819 10.89410412 18.22962775]

New x_best found: [18.04886081 45.2830216 11.87113351]

--- Iteration 4/100 ---

Best fitness so far: 22.794

x_best: [18.04886081 45.2830216 11.87113351]

--- Iteration 5/100 ---

Best fitness so far: 22.794

x_best: [18.04886081 45.2830216 11.87113351]

--- Iteration 6/100 ---

Best fitness so far: 22.794

x_best: [18.04886081 45.2830216 11.87113351]

--- Iteration 7/100 ---

Best fitness so far: 22.794

x_best: [18.04886081 45.2830216 11.87113351]

--- Iteration 8/100 ---

Best fitness so far: 22.794

x_best: [18.04886081 45.2830216 11.87113351]

--- Iteration 9/100 ---

Best fitness so far: 22.794

x_best: [18.04886081 45.2830216 11.87113351]

--- Iteration 10/100 ---

Best fitness so far: 22.794

x_best: [18.04886081 45.2830216 11.87113351]

--- Iteration 11/100 ---

Best fitness so far: 22.794

x_best: [18.04886081 45.2830216 11.87113351]

New x_best found: [42.7174555 41.8353711 15.28995203]

--- Iteration 12/100 ---

Best fitness so far: 3.292

x_best: [42.7174555 41.8353711 15.28995203]

--- Iteration 13/100 ---

Best fitness so far: 3.292

x_best: [42.7174555 41.8353711 15.28995203]

--- Iteration 14/100 ---

Best fitness so far: 3.292

x_best: [42.7174555 41.8353711 15.28995203]

New x_best found: [42.17920593 41.90357824 15.03375838]

--- Iteration 15/100 ---

Best fitness so far: 2.894

x_best: [42.17920593 41.90357824 15.03375838]

--- Iteration 16/100 ---

Best fitness so far: 2.894

x_best: [42.17920593 41.90357824 15.03375838]

--- Iteration 17/100 ---

Best fitness so far: 2.894

x_best: [42.17920593 41.90357824 15.03375838]

--- Iteration 18/100 ---

Best fitness so far: 2.894

x_best: [42.17920593 41.90357824 15.03375838]

--- Iteration 19/100 ---

Best fitness so far: 2.894

x_best: [42.17920593 41.90357824 15.03375838]

--- Iteration 20/100 ---

Best fitness so far: 2.894

x_best: [42.17920593 41.90357824 15.03375838]

--- Iteration 21/100 ---

Best fitness so far: 2.894

x_best: [42.17920593 41.90357824 15.03375838]

--- Iteration 22/100 ---

Best fitness so far: 2.894

x_best: [42.17920593 41.90357824 15.03375838]

New x_best found: [39.23316963 41.94360999 14.61999157]

--- Iteration 23/100 ---

Best fitness so far: 2.124

x_best: [39.23316963 41.94360999 14.61999157]

--- Iteration 24/100 ---

Best fitness so far: 2.124

x_best: [39.23316963 41.94360999 14.61999157]

New x_best found: [39.7664325 41.94219643 14.82678059]

--- Iteration 25/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 26/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 27/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 28/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 29/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 30/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 31/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 32/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 33/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 34/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 35/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 36/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 37/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 38/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 39/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 40/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 41/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 42/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 43/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 44/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 45/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 46/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 47/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 48/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 49/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 50/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 51/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 52/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 53/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 54/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 55/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 56/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 57/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 58/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 59/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 60/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 61/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 62/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 63/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 64/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 65/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 66/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 67/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 68/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 69/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 70/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 71/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 72/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 73/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 74/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 75/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 76/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 77/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 78/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 79/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 80/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 81/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 82/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 83/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 84/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 85/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 86/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 87/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 88/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 89/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]

--- Iteration 90/100 ---

Best fitness so far: 1.964

x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 91/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 92/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 93/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 94/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 95/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 96/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 97/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 98/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 99/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]
--- Iteration 100/100 ---
Best fitness so far: 1.964
x_best: [39.7664325 41.94219643 14.82678059]

=== Algorithm complete. Landing all drones. ===

Sensors stopped.

Log saved to logs/run_20260625_045933.json

Final best fitness: 1.964

Final best position: [39.7664325 41.94219643 14.82678059]